Assignment 2: Experiment Comparison
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This assignment contained 4 experiments:
1. Data set version 1 (limited features) with hyperparameters n_estimators = 250 and max_depth = 5
2. Data set version 1 (limited features) with hyperparameters n_estimators = 500 and max_depth = 10
3. Data set version 2 (expanded features) with hyperparameters n_estimators = 250 and max_depth = 5
4. Data set version 2 (expanded features) with hyperparameters n_estimators = 500 and max_depth = 10

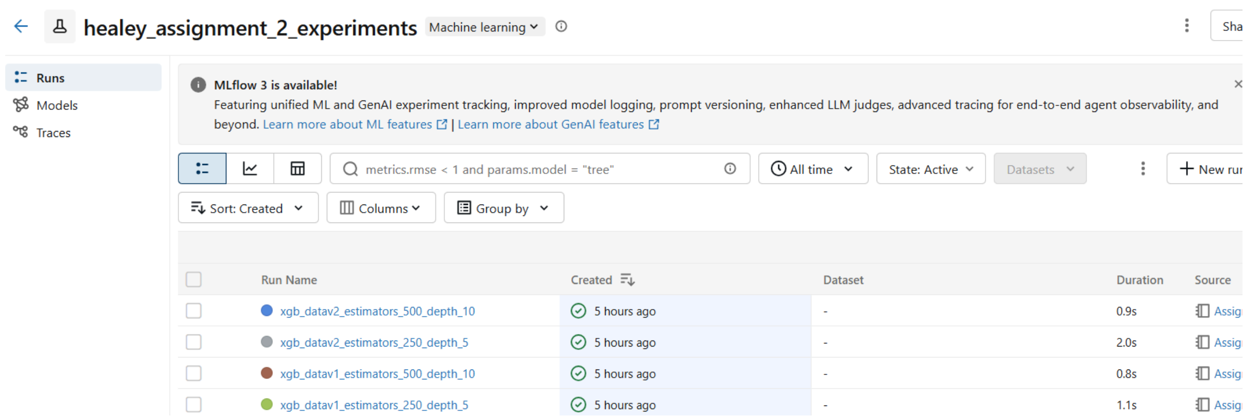
A priori, I expect experiment 3 to have the best evaluation metrics, because it has the most robust set of features and its hyperparameters allow for learning without encouraging overfitting (like the larger set of hyperparameters might do). However, it is possible that experiment 4 has more ideal hyperparameters and could outperform experiment number 3. Overall, I do not expect experiments 1 or 2 to perform the best, since they have fewer explanatory variables.

Now, let's evaluate my hypothesis and the results. All results are pulled from MLflow.

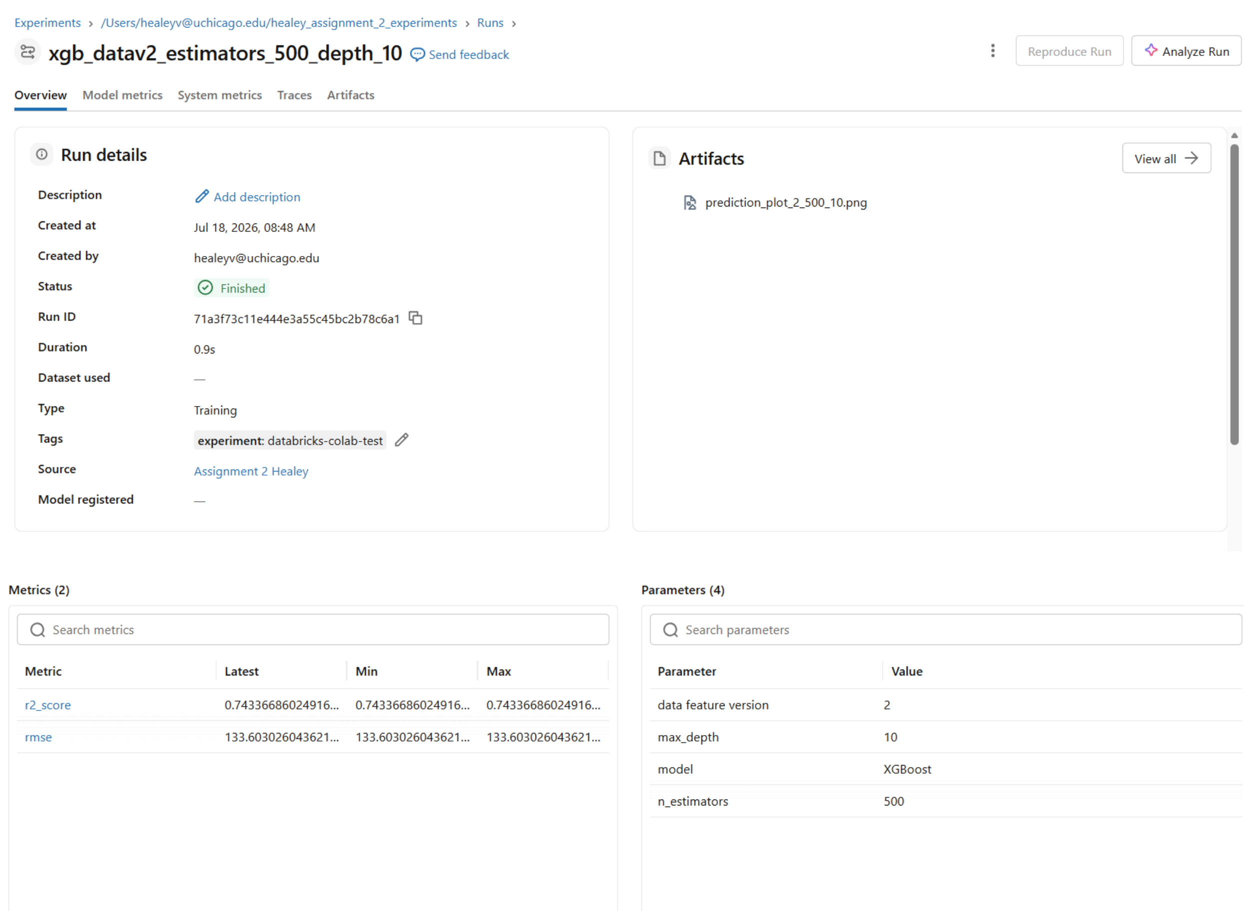
Experiment	R2	RMSE
1	.77	126.40
2	.74	134.24
3	.77	126.24
4	.74	133.60

We see that experiment 3 had the highest R2 and lowest RMSE, suggesting it was the best out of the four trials. Notably, though, it only barely outperformed experiment 1, which had the same hyperparameters but slightly fewer features. This suggests that the additional features created in data version 2 had a relatively limited benefit on the final model. Instead, the hyperparameter configuration appears to be more important to the results.

See page 2 for screenshots documenting the MLflow environment.



MLflow environment in DataBricks with model results



Specific model results (including artifacts, metrics, and parameters) in MLflow.